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Reverse Engineering Course

Dynamic Analysis

Barak Gonen

Dynamic Analysis

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- Easier to understand code functionality
  - Check values in registers, memory
  - Check stack: arguments and return addresses
- Virustotal

SimpleKey

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- Crack SimpleKey.exe
- Jump to address – hex view - G
- https://data.cyber.org.il/reversing/SimpleKey.exe

Dynamic Analysis

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- Modify on the fly:
  - Registers
  - Code
  - Data
- Patch executable

TeaParty

- Modify registers to drink your tea
- https://data.cyber.org.il/reversing/TeaParty.zip

Change Code

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- Edit -> Patch program -> Change byte
- Edit -> Patch program -> Assemble
- Use “Rename” memory address for jumps
- Apply changes to executable:
  - “Apply patches to input file”

IDA – Play with Patching

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- https://data.cyber.org.il/reversing/BreakMe.exe
- Patch and apply patches to BreakMe.exe:
  - Step 1: Accept all but correct password
  - Step 2: Accept all passwords
  - Step 3: Don’t even ask for a password
    - Hint: Rename memory location

Home Work

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- Patch InjectMe.exe
- Tip: study msdn WinMain
- https://data.cyber.org.il/reversing/InjectMe.exe

Patch Data

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- Vegas
- Reach good message:
  - No code patching
  - No arguments

Hunt the flag

- https://data.cyber.org.il/reversing/ShabakFirst.exe